DAVID NYARKO

AI and Machine Learning Robotics Systems Engineer

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Baltimore, Maryland USA

SUMMARY

Passionate Software Engineer with a strong background in machine learning infrastructure, neural network scaling, and AI deployment. Skilled in building and optimizing AI pipelines, distributed training frameworks, and tools for visualizing model performance metrics using Python and C++. Proficient in system-level software and hardware-software interactions, with hands-on experience in deploying deep learning models and optimizing GPU clusters for large-scale training. Demonstrated ability to manage and analyze complex datasets and coordinate with hardware teams to maintain high availability and throughput for machine learning jobs. Excited to contribute to Tesla's Autonomy team by enhancing AI infrastructure and advancing autonomous driving technology through innovative software solutions.

EXPERIENCE

AI Systems Engineering Research Fellow

Data Engineering and Predictive Analytics Lab

01/2023 - Present Maryland, USA

- Part of a team that design and implement an Autonomous Wheelchair with Phase 3 of the project being tested at the Baltimore/Washington International Thurgood Marshall Airport **(Project Lead)**
- Built and Published ELT: Easy Label Trainer, an object detection and segmentation model for the computer vision community
- Creator and maintainer of **edge-tpu-silva**, a Python open-source library that helps run large models on Edge Computers
- Build text recognition models on video streams with optical character recognition

Machine Learning Engineer

SETI Institute

05/2023 - 08/2023 Califonia, USA

- Increased accuracy by utilizing advanced computer vision models and about a million synthetic image data for training
- Enhanced system reliability, increasing uptime by 25%, through implementing ensemble methods with diverse data sources
- Led efforts in data preprocessing and augmentation, seamlessly integrating synthetic and real data to optimize model performance for practical applications
- Collaborated closely with domain experts to ensure real-world relevance, rigorously testing models on real data to validate their effectiveness
- Presented research findings and progress updates to SETI Institute teams and the US Department of Energy, highlighting the successful utilisation of synthetic data and real data testing in achieving project goals

Drone Technology Specialist

AirForce-National Reconnaissance Office STEM Academy

04/2023 - 06/2023 Maryland, USA

- Created and instructed courses on autonomous drone programming and integration
- Provided guidance and coaching to students in the realm of electronics circuit design and the development of rockets

Graduate Teaching Assistant

Morgan State University (MSU)

08/2022 - 01/2023 Maryland, USA

- Assisted in an Introduction to C Programming for an undergraduate level 100 class under the School of Graduate Studies Graduate Assistant contract award
- Assisted Discrete Systems class, an undergraduate level 300 class

SKILLS

Deep Learning Algorithms | Blender | Arduino|ESP Boards | C/C# programming | Computer Vision | Dart | ROS1|ROS2 | Data Engineering | Flutter | GCP|AWS|APPWRITE | Github | Gitlab | IoT | Matlab | Microcontrollers | Predictive Analytics | Python | Raspberry PI | Robotics | Scikit | Tensorflow | Unity 3D | Video Editing | 3D Printing | React Native | React|Typescript | Nvidia Isaac Sim | Realistic 3D Modeling | Nvidia Jetson | LLMs | Foundation Models | Digital Twins | VR/AR Simulations

EDUCATION

Doctor of Engineering

Morgan State University

08/2022 - Present
Baltimore, MD. USA

- Electrical and Computer Engineering
- (3rd Year)

GPA **4.0** / 4.0

B.Sc. Engineering Physics

University of Cape Coast

07/2017 - 12/2021
Cape Coast, Ghana

GPA **3.64** / 4.0

General Science

Mampong Presby Senior High School

08/2014 - 05/2017 Akuapem, Ghana

EXPERIENCE

Engineering Lead

SwiftDynamics

06/2020 - 06/2024 Accra, Ghana

- Design and development of a Robot Operating System for SwiftDynamics AI which employs machine learning and artificial intelligence
- Design and construction of Africa's first AI social robot called Claby to assist persons with disabilities (The first robot to use the swift dynamics robot operating system)
- Design and construction of a portable ventilator using Arduino micro-controllers to support the fight against COVID-19 in my region. (University of Cape Coast)
- Part of the design and construction of a weather station for the Ghana Space Science Agency
- Developer of "James Town", a photorealistic game that features Africa's culture and history

Engineering Intern

Ghana Atomic Energy Commission (GAEC)

05/2019 - 08/2019 Accra, Ghana

- Improved attendance accuracy by 95% by designing a facial recognition system with Raspberry Pi
- Led design of detachable touchless dispenser, resulting in a cost reduction of 45% for production
- Member of a team that designed and implemented an Arduino-based single-stack Traffic Light
- Member of a team that designed and implemented an Arduino-based automatic irrigation system with an IOT APP

PUBLICATIONS

ELT: Easy Label Trainer

Springer Nature, Lecture Notes in Networks and Systems

David Nyarko, Kelechi Nwachukwu, Anjolie Anthony, Kofi Nyarko

07/2024

This paper presents ELT-Easy Label Trainer (ELT), a tool designed to automate and streamline the process of labeling and training object detection models. It integrates advanced models like YOLOv8, YOLOv9, and YOLO-World for robust object detection and segmentation using descriptive text prompts. ELT employs pseudo-labeling techniques to enhance detection accuracy, even for objects outside standard datasets. By combining these models with automated tracking methods, ELT significantly eliminates manual labeling effort and improves model performance, demonstrating its effectiveness on the GMOT-40 benchmark for multiple object tracking.

STRENGTHS

Teamwork and Problem Solving

Successfully collaborated in a diverse, randomly assigned team at the Emory Health AI Datathon to provide solutions within 48 hours. Created vector embeddings for X-rays and analyzed patterns of missing information in the Social Determinants of Health (SDOH) dataset, uncovering how socio-economic factors impact health outcomes and revealing biases in healthcare data.

Innovation in AI and Robotics

Pioneered the creation of the edge-tpu-silva Python library, enabling efficient execution of large models on edge devices for computer vision tasks. Developed innovative AI-driven robotic systems, such as Africa's first AI social robot, Claby, and an automated portable ventilator, demonstrating a strong capacity for translating AI research into impactful real-world applications.

VOLUNTEERING

Drone Technology Specialist

AirForce-National Reconnaissance Office STEM Academy

04/2023 - 06/2023 Maryland, USA

Created and instructed courses on autonomous drone programming and integration

Provided guidance and coaching to students in the realm of electronics circuit design and the development of rockets

Executive Member

Center for Equitable Artificial Intelligence and Machine Learning Systems - K-12 AI Mentoring Program

08/2023 - Present Maryland, USA

Organized AI and ML workshops for high school students at local libraries.

Collaborated with a team of AI professionals to develop accessible learning materials, inspiring the next generation of AI enthusiasts and promoting diversity in STEM fields.

AWARDS

2nd Place WINNER, My team won second place out of 12 teams at the Emory Health AI Datathon 2024

Certificate of Participation, Awarded for contribution to the 2024 Health AI Summer School, Symposium and Datathon at Emory University, 2024

CERTIFICATE OF ACHIEVEMENT, Awarded for exceptional performance and outstanding achievements in applied Artificial Intelligence and Machine Learning, advancing science for FDL 2023 and the US Department of Energy. SETI Institute's FDL Program 2023

WINNER, Best student in Engineering in Ghana. Ghana Tertiary Awards 2020

WINNER, Rising star in embedded programming. PHYSAG Excellence Awards 2021

WINNER, Most Innovative Engineering Physics Student. UCC, School of Physical Sciences Dean's Award 2017